# Inquiry-Relative Categories: English Interjections as a Worked Case

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Abstract

Disputes about linguistic categories often combine different questions about the same expressions. English interjections provide a worked case: grammatical distribution, expressive meaning, and sequential use motivate overlapping classifications whose boundaries needn't coincide. I develop an account of inquiry-relative category standing in which each proposal specifies its bearers, grouping basis, intended contribution, scope, and alternatives. The three classifications illustrate an open family. Phonological organization and historical ancestry motivate further candidates; social regulation and processing connect the traditional interjection domain to broader classes. Classifications of realizations and historical changes require explicit relations to classifications of sign-types. These distinctions explain how category disputes can be separated without making classification arbitrary or assigning automatic priority to a discipline. Documented recruitment histories support particular explanations, while audits of two corpus studies withdraw their intended confirmatory interpretations. The resulting framework distinguishes descriptive organization, projection, representational benefit, and causal explanation as contributions requiring appropriate evidence. Overlap, gradience, and shared mechanisms are compatible with justified classification; the number and reach of useful categories remain outcomes of inquiry.

Keywords: inquiry-relative categories; category pluralism; projectibility; interjections; sign-types; path dependence

AI USE. The large language models Claude Opus 4.8, Claude Opus 5, Claude Opus 5.5, GPT-5.6 Sol, GPT-6 Astra Pro, and GPT-6 (Codex) served as drafting and editing aids throughout the preparation of this paper. I am responsible for all theoretical claims, arguments, errors, and interpretive choices.

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# 1 Introduction

Standalone *Damn!*, complement-taking *Damn these mosquitoes!*, and prenominal *damn* in *the damn car* present different combinations of grammatical and semantic properties. The modifier participates in a noun phrase while retaining an expressive contribution. The standalone form can constitute a stance display. The complement-taking expression has internal structure even when the whole expression stands apart from a surrounding clause. A single judgment about whether *damn* is an interjection leaves these differences unresolved.

Which judgment matters depends on the inquiry. A grammarian investigating phrase structure needs a distinction that a semanticist studying expressive contribution may properly cross. An analyst of conversation asks a further question about conventional actions and recipients' responses. The difficulty arises when evidence answering one question is taken to settle all three, or when failure to share one extension is treated as failure to identify any defensible category.

This paper develops an account of INQUIRY-RELATIVE CLASSIFICATION. Its central claim is that a traditional linguistic domain can contain several overlapping classifications, each assessed for the distinctions, inferences, or explanations it supports. An inquiry identifies a question and a standard of success. Evidence constrains which grouping answers it. The same objects can participate in several justified classifications without one inquiry acquiring general priority over the others.

English interjections supply the worked case. I develop a grammar-facing $\text{INTERJECTION}_{\text{SYN}}$, a semantics-facing $\text{INTERJECTION}_{\text{SEM}}$, and an interaction-facing $\text{INTERJECTION}_{\text{PRAG}}$. These are analytical reconstructions drawing on established traditions, rather than three technical senses already distinguished consistently in the literature. They share conventionalized sign-types as bearers: identifiable expressions in conventional uses. The three uses of *damn* above belong to one historical family but require separate analysis at this grain.

The three classifications are examples within an open family. Phonological organization can motivate another grouping of sign-types, and historical ancestry can relate descendants with different present distributions. Other inquiries classify realizations, developmental episodes, or processes of change. Their connection to sign-types needs to be stated. These extensions show why the subscript in $\text{INTERJECTION}_{\text{X}}$ identifies a particular inquiry, rather than allocating one category to each disciplinary department.

The general proposal has two constraints. First, a candidate specifies its bearer, grouping basis, intended contribution, scope, and relevant alternatives. Second, the evidence supporting one contribution isn't automatically evidence for another. A useful descriptive contrast, a successful prediction, an economical representation, and a supported causal explanation can come apart. The standards apply equally to the initial three classifications and to further candidates.

The problem is familiar in part from the distinction between grammatical clause type and speech act. An interrogative can perform a request, while an answer-seeking action can be performed without interrogative form. The proper-noun/proper-name distinction offers another comparison: a name can be headed by a common noun. The interjection case develops the general problem further because several traditions compete for the same label, and their diagnoses mix properties of forms, uses, occurrences, and histories.

The historical argument concerns PATH DEPENDENCE: an expression's development can leave source-specific traces after its use converges with that of other expressions. Gehweiler's history of *gee* supplies a documented case; the comparison with complement-taking *damn* is partly reconstructive.

These histories help distinguish an explanatory classification by ancestry from a classification by current distribution. They also show why a present trace needs independent historical evidence before it can count as support for the proposed route.

The corpus material has a narrower role. Geographic frequency differences and changes in tagged inventories provide examples of what a measure can and can't establish about a classification. Audits invalidate the intended confirmatory interpretations of two further analyses, of derivatives and *fie*; the appendices preserve their records and explain the failures. The paper uses those failures to delimit its evidence rather than treating precise numerical outputs as grounds for category standing.

Section 2 develops the three classifications through concrete contrasts. Section 3 explains their common assessment standard. Sections 4 and 5 develop phonological and historical extensions, including the relations among different bearers. Section 6 considers broader social and processing classes. The empirical assessment and discussion then establish the limits and consequences of the general account.

## 2 Three classifications of conventional signs

Grammars and studies of meaning and interaction have organized the traditional interjection domain differently. Jespersen (1924, p. 90) treated interjection as a manner in which words of other categories are used. *The Cambridge Grammar of the English Language* (*CGEL*) recognizes a lexical category (Huddleston & Pullum, 2002, p. 22). The *Longman Grammar of Spoken and Written English* places interjections within the broader class of inserts, while assigning *damn*, *gosh*, and *gee* to expletives (Biber et al., 1999, pp. 1082–1083, 1094–1095).

The disputes concern the organization of a well-studied domain. Libert (2019) questions the common description of interjections as neglected; Cram (2008, p. 63) identifies their recurring role in controversies over the syntax–pragmatics and semantics–pragmatics boundaries. A comparison of classifications should preserve the contrasts those controversies disclose. It should also distinguish an alternative grouping from a disagreement about the facts.

### 2.1 Grammar and expressive meaning

The grammar-facing classification begins with standalone and supplement distribution. In *Damn, we're going to be late!*, *damn* is appended to the clause without being its complement or modifier. *CGEL* treats interjections as occurring on their own or functioning as supplements (Huddleston & Pullum, 2002, pp. 1350, 1361). In their conventional interjectional uses they're characteristically invariant and resist the ordinary constituent positions available to nouns, verbs, or adjectives.

Supplement function is shared by several categories. Compare the interjection with the relative clause and noun phrase in the following examples, adapted from *CGEL* (Huddleston & Pullum, 2002, pp. 1356–1357, 1360).

(1) a. *Ah, so you were there after all!*
b. *We called in to see Sue's parents, which made us rather late.*
c. *A university lecturer, Dr Brown, was arrested for the crime.*

Their external relation to a host doesn't erase their internal grammatical differences. A nominal supplement remains a noun phrase; a relative clause has clause structure. For the same reason, the external detachment of *Damn these mosquitoes!* is compatible with internal complementation.

*CGEL* tentatively treats such cases as exceptional interjections heading an interjection phrase with a noun-phrase complement (Huddleston & Pullum, 2002, 1361 n. 70). A verbal analysis remains a substantive alternative. The complement alone doesn't decide between them.

Prenominal *damn* provides a different contrast. In *the damn car*, it occurs inside the noun phrase, with the phrase's prosodic packaging, while contributing an attitude toward its referent. Potts (2007, pp. 171, 182) distinguishes that expressive contribution from descriptive content. The expression can preserve this semantic distinction while departing from the standalone distribution characteristic of grammatical interjections.

INTERJECTION$_{\text{SEM}}$ develops this expressive classification. EXPRESSIVE here concerns a conventional contribution distinguishable from descriptive content; the associated USE-CONDITIONAL analysis concerns what using the expression conveys. Potts (2007, pp. 166–167) identifies several diagnostics, including perspective dependence, immediacy, and intensifying repetition. Their applicability to a proposed member is an empirical question. Mere independence from the host proposition is insufficient, since a supplement can contribute a descriptive proposition of its own.

The two *damn* uses make the benefit of separate classification visible. The grammatical analysis preserves a difference in distribution; the semantic analysis preserves a common kind of contribution. Averaging these observations into a single degree of interjectionhood would discard which respect is shared and which differs. A constructional inventory could also record both relations. The proposal concerns their organization as distinct classifications and the consequences attributed to them.

## 2.2 Interaction and its subtypes

An interaction-facing classification groups conventional minimal resources by their availability for identifiable jobs in sequences of action. Continuers such as *mm-hm*, repair initiators such as *huh?*, and change-of-state uses of *oh* are central cases in accounts of interjections in conversation (Dingemanse, 2024, pp. 257–259, 263–265). Their importance becomes less visible if the entire domain is organized around emotional exclamation.

A continuer can support another speaker's ongoing turn. A repair initiator can make a response to a problem of hearing or understanding relevant. Those are different actions with different consequences. Assigning both to INTERJECTION$_{\text{PRAG}}$ preserves their availability as conventional resources; it doesn't predict the same next turn for both. The evidential question for the superordinate class concerns what generalization it supports beyond the separate descriptions of its subtypes.

The selected continuer and repair uses also show why an interactional contribution needn't be an expressive one. A speaker can convey an attitude through a particular *mm-hm* or *huh?* without that attitude being the conventional expressive contribution of the use. I provisionally exclude the selected readings from INTERJECTION$_{\text{SEM}}$, while assigning them to INTERJECTION$_{\text{SYN}}$ on their conventional standalone distribution and to INTERJECTION$_{\text{PRAG}}$ on their sequential role. Evidence for another reading could warrant another assignment.

Routine formulae expose disagreement within this interactional inquiry. Ameka (1992, pp. 108–110) separates forms such as *bye*, *sorry*, and *thank you* from interjections, emphasizing addressees, conventional social situations, and speech-act status. Wilkins (1992, p. 142) treats them as a pragmatic and semantic subtype. Both attend to meaning and interaction. Allocating one author to grammar and the other to pragmatics would misdescribe the disagreement.

Under the criterion adopted here, closing *bye* belongs to INTERJECTION$_{\text{PRAG}}$: it's a conventional minimal resource in a recognizable sequential environment. Closing formulae remain distinguishable

from continuers, repair initiators, and stance displays. This broad assignment is a proposal whose usefulness can be assessed against narrower groupings. An inquiry can support several classifications, including subtypes, rather than requiring one extension for all interactional purposes.

### 2.3 The common bearer and the contrasts it preserves

A sign-type is an identifiable expression in a conventional use. Its profile is the combination of grammatical, semantic, interactional, and other properties attributed to it. Profile is a descriptive term here. The use can retain its identity while a proposed property turns out to be absent. Otherwise every surprising observation would identify a different object and leave the original classification immune to correction.

Different conventional uses require evidence for their individuation. The recurrent contrast between standalone and prenominal *damn* supports distinguishing them. A spelling alone doesn't: a word family can include several signs, and a sign can have spelling variants. Table 1 records the selected uses and the present analytical assignments. They're neither counts of occurrences nor results of a new annotation study.

Table 1: Selected uses under the three worked classifications. A check marks assignment, a dash exclusion under the stated analysis, and a question mark an unresolved assignment. The prenominal row concerns *damn* as a modifier within the noun phrase.

| Selected use | $\text{Intj}_{\text{syn}}$ | $\text{Intj}_{\text{sem}}$ | $\text{Intj}_{\text{prag}}$ |
|---|---|---|---|
| *gee!* | ✓ | ✓ | ✓ |
| *damn!* (stance) | ✓ | ✓ | ✓ |
| *damn these mosquitoes!* | ? | ✓ | ? |
| Prenominal *damn* | – | ✓ | – |
| *mm-hm* (continuer) | ✓ | – | ✓ |
| *huh?* (repair) | ✓ | – | ✓ |
| *bye* (closing) | ✓ | – | ✓ |
| *Bang!* (depiction) | ? | – | ? |

The depictive use of *Bang!* illustrates another separation. Standalone distribution makes it a candidate for $\text{interjection}_{\text{syn}}$, but that distribution alone doesn't distinguish an interjection from a nominal fragment. Its grammatical assignment remains open here. Depicting a sound supplies a different semantic relation from the expressive contribution of *damn*. Work on ideophones and onomatopoeia makes that difference consequential (Dingemanse, 2018; Dingemanse & Akita, 2017; Meinard, 2015). Its conventional sequential role remains a further question. Standalone packaging by itself doesn't answer it.

The unresolved cases matter as much as the positive assignments. Complement-taking *damn* raises a grammatical question independently of whether it has expressive meaning. Its absence from a dataset of backchannels would say little about its availability for stance displays. A question mark preserves that separation. Treating missing evidence as exclusion would make the apparent distance between classifications partly an artefact of what investigators happened to study.

These comparisons establish a reason to keep the judgments distinct. They leave open how much further work each grouping performs. A grammatical category may support additional distributional

expectations; the expressive classification may extend to signs never called grammatical interjections; interactional subtypes may be more informative than their common superordinate class. Those possibilities require one assessment standard that applies without privileging any of the three.

## 3 What justifies an inquiry-relative classification?

An inquiry gives an investigator a reason to distinguish cases. Evidence then determines whether a proposed grouping supports that distinction. For an expressive modifier such as *damn*, the grammatical question concerns its place inside the noun phrase; the semantic question concerns its contribution to interpretation. A useful answer to either can leave the other open. Calling both answers classifications doesn't give them equal evidence, but it gives them the same obligation to state what they accomplish.

### 3.1 Bearers, grouping, contribution, and alternatives

Four questions organize that obligation. What bears membership? What groups the members? What does membership help an investigator distinguish, infer, or explain? And what alternative would serve the same inquiry? Each proposal also needs a scope: a language or community, relevant conditions, and, where the claim concerns change, a period. These questions apply to the three worked classifications and to every proposed extension of the family.

The bearer is the object being classified. In the worked cases it's a conventionalized sign-type, an expression in an identifiable conventional use. A phonologist may instead classify its realizations, and a historian may classify the changes linking it to earlier signs. A statement about the frequency of realizations doesn't immediately become a statement about the number of types or the rate of change. Specifying the bearer prevents those transfers before they enter an argument.

The grouping basis is the evidence or relation used to assign members. Grammatical distribution, conventional expressive contribution, common ancestry, and participation in a socially recognized repertoire are different possibilities. A grouping can be categorical or graded; it can be supported by one diagnostic or several. Its shape should follow the evidence relevant to the inquiry. A demand for a unique mechanism or a sharp boundary would exclude proposals before their intended contribution had been assessed.

The contribution concerns what investigators can do with the grouping. A descriptive classification may organize recurrent contrasts across cases; a projective one supports expectations about cases or properties beyond those used to assign membership; an explanatory one connects differences to independently supported processes. These achievements have different requirements. A label that restates its admission criterion has identified a set, but its further scientific value remains to be shown.

An alternative can be a broader category, a narrower subtype, a continuous dimension, or a different organization of the same evidence. For expressive *damn*, an undifferentiated interjection category competes with a classification that groups it with other expressives despite their grammatical differences. For recipient responses, a broad interactional class competes with separate action subtypes. The appropriate comparison depends on the question; a smaller vocabulary isn't invariably a better account.

Table 2 shows how the questions constrain a proposed classification. The final column identifies what would still require evidence. The example concerns expressive contribution rather than a claim that all signs in the traditional interjection domain share it.

Table 2: Assessing the expressive classification

| Question | Proposed answer | Evidential consequence |
|---|---|---|
| Bearer | Conventional expressive uses, including standalone and prenominal *damn* | Establish the uses independently of the particular target being assessed. |
| Grouping | A conventional contribution distinguishable from descriptive content | Host-proposition preservation alone is insufficient; descriptive supplements can preserve it too. |
| Contribution | Preserve a semantic generalization across different grammatical constructions; investigate further expressive diagnostics | Distinguish an attested contrast from a prediction about an unexamined sign or environment. |
| Alternative | An undifferentiated interjection category, item-specific descriptions, or a broader expressive classification | Compare their coverage and consequences for the semantic question, with comparable evidence. |

### 3.2 From candidate to supported classification

A candidate earns consideration by making those questions answerable. A supported classification has evidence for the contribution it claims. Comparative merit requires a further judgment about alternatives. The existing grammatical, semantic, and interactional traditions supply substantial descriptive resources, but the three reconstructions in this paper inherit their evidence selectively. Their shared notation supplies no evidence of its own.

The distinction matters when extending the family. A phonological stratum with a clear bearer and plausible generalizations deserves investigation before those generalizations have been demonstrated. Its status at that point is prospective. The same is true of an untested projection from $\textsc{interjection}_{\textsc{syn}}$ or $\textsc{interjection}_{\textsc{prag}}$. Equal standards of assessment permit unequal evidential support; they require that the difference in support be made visible.

Even a well-supported classification has a limited reach. Evidence for the expressive contribution of prenominal *damn* doesn't settle the contribution of repair *huh?*. An analysis of one sequential action doesn't establish a common next-turn prediction for all interactional resources. Generalization requires evidence for the grouping at the breadth claimed. Where only a subtype is informative, the larger category may remain useful for description without acquiring that subtype's predictions.

### 3.3 Projection and identity

Projectibility concerns warranted expectations beyond the cases in hand (Goodman, 1955). A projective claim specifies what's been observed, what's expected, and over which bearers and conditions. Its warrant is the evidence supporting that expectation. A precise specification can have little warrant; a familiar association can be useful even while its cause remains unknown. These distinctions separate the content of a claim from its evidential and explanatory status (Reynolds, 2026c, pp. 11–12).

Suppose independently assessed source contexts identify an expressive use, and the target is its

behaviour under repetition in another context. Repetition can't also have been the case-level evidence used to admit that use to the source class. Potts's expressive diagnostics provide possible targets of this kind (Potts, 2007, pp. 166–167). Their availability in a theory doesn't establish their success for every proposed member.

The same separation constrains sign-type identity. Standalone and prenominal *damn* can be distinguished through recurrent constructional and interpretive contrasts. But a target failure can't automatically create a new sign-type. If determiner resistance is the prediction, encountering a determiner doesn't by itself license excluding the case as conversion. The identification rule has to be fixed from source evidence; an independently supported change in that rule creates a revised claim requiring fresh assessment.

An informative restriction also needs relevant opportunities. Failure to observe an inflected form says little if the sampled environments wouldn't elicit inflection under a competing analysis. For a grammatical projection, the design should name the source construction, the target construction, and the competing expectation. For an interactional one, it should identify the action and the environments in which recipients' treatment would count as support or counterevidence. Such specifications are needed for a test; the classifications themselves don't supply them.

### 3.4 Representation and explanation

A category can organize evidence without adding a new fact to a complete description of its members. If its label is computed deterministically from fully observed features, adding that label to those same features adds no information. The useful comparison concerns an investigator with limited, noisy, or incomplete observations. Grouping can support pooling across cases, reduce the number of separate assumptions, or represent dependencies that an alternative overlooks. Those benefits are assessable for a specified task.

The intersection of several categories deserves the same assessment. Joint grammatical, expressive, and interactional membership may be an economical grouping for some questions. Its derivability from the separate memberships neither establishes nor defeats that possibility. Conversely, an advantage over an artificially restricted additive model wouldn't establish the intersection as a privileged category for every inquiry. The comparison should identify what the competing representations already contain.

Historical explanation poses a different question. If a documented recruitment process produced a present property, controlling for that property can remove the association through which the process is recognized. The disappearance of an additional predictive effect then leaves the historical explanation open. A test of prediction from limited present-day evidence and a test of the proposed history have different evidential requirements. Section 5 develops the distinction through *gee*.

Processing and social explanations require similar care about which variables are compared. Repeated use could help produce fluent retrieval. Adjusting for measured frequency asks what association remains conditional on that measure; it doesn't by itself identify repetition's total or direct causal contribution. A social classification may concern audience-sensitive selection, making the relation to audience part of the target rather than a nuisance to eliminate.

These explanations also differ in strength. A recurring profile can be stable while its causal ordering remains unknown. An independently supported historical dependence can be real without demonstrating that the whole profile recurs across an inventory. Maintenance requires evidence that a process preserves a profile, and corrective control requires a response to disturbance. I reserve

homeostatic for that stronger control claim, following Reynolds (2026b), a narrower use than the one in Boyd (1999, p. 143). No such control claim is established here.

Inquiry has a limited and substantive role. It selects the bearer, the contrast of interest, the scope, and the standard of success. Observed distributions and supported explanations constrain which groupings meet that standard. This permits several justified classifications without making category standing a consequence of an investigator's preference.

## 4 Phonological organization and situated realization

Phonological evidence already enters descriptions of interjections in two forms. Some signs permit sounds or combinations unusual elsewhere in a variety; particular occurrences also have distinctive prosodic packaging. These observations motivate different classifications. The first can group sign-types by their phonological possibilities. The second groups realizations by how speakers produce them in context. Their relation is empirically interesting precisely because they don't classify the same objects.

### 4.1 A phonological stratum of signs

English conversational clicks make the first possibility concrete. The click represented by *tut-tut* differs from an ordinary spelling pronunciation (Wilkins, 1992, p. 147). Forms such as *ugh* permit sounds outside the usual phoneme inventory of standard British and American English (Quirk et al., 1985, p. 853). A phonological inquiry can ask which conventional signs admit such resources, how their variants are organized, and how speakers acquire or adapt them.

Ameka (1992, pp. 112–113) explicitly distinguishes a phonological class of unusual forms from a grammatical category: onomatopoeic words can share the relevant phonological properties while belonging to different grammatical categories. A proposed phonological stratum would cross the grammatical boundary. Its grouping basis might concern permitted segments, syllable structures, or variation patterns. These are several possible inquiries, whose extensions needn't coincide with one another or with $\textsc{interjection}_{\textsc{syn}}$.

An account of the stratum would gain projective reach if membership organized acquisition, adaptation, or alternation beyond the diagnostic used to identify it. For example, a classification based on independently established segmental possibilities could be assessed against patterns of spelling pronunciation. That's a prospective comparison. The existence of unusual sounds and reported alternatives establishes a descriptive motivation; it doesn't yet show one phonological class with a uniform learning or adaptation profile.

Regular phonology matters too. In a ten-language study of open repair interjections, Dingemanse et al. (2013) find broad phonetic similarities alongside language-specific vowel targets and intonation. The repair interjections in Icelandic and Cha'palaa have falling contours, consistent with local interrogative prosody. The study concerns comparable repair environments. It supplies no reason to assign every occurrence resembling *huh* to one phonologically exceptional class.

Phonological classification thus has work to do within ordinary linguistic organization as well as at its margins. An inquiry into repair intonation could group forms with interrogative resources; another into unusual segments could group conversational clicks with other marked forms. Both could be useful. A single label such as $\textsc{interjection}_{\textsc{phon}}$ would obscure the difference unless the inquiry and grouping basis were specified.

### 4.2 Prosodic classes of realizations

A prosodic boundary is a property of a realization. A sign-type can characteristically occupy its own intonation phrase while allowing different realizations in different environments. Conversely, realizations of grammatically different signs can share a boundary pattern. Supplements provide a familiar example: interjections, noun phrases, and relative clauses can all be prosodically detached (Huddleston & Pullum, 2002, pp. 66, 1350).

The relation can be stated in two steps. First identify an occurrence as a realization of a sign-type using evidence appropriate to that identification. Then classify its prosodic organization from the acoustic record. A claim about the sign-type concerns the distribution of those realizations over relevant environments. One detached occurrence is evidence about a possibility; an adequately sampled distribution is needed for a claim about what's characteristic.

Interaction supplies some of those environments. Ogden (2013) describes English clicks and percussives in stance-taking, sequence management, word searches, and turn timing, with visual behaviour contributing to their interpretation. A classification of productions by timing and articulation can intersect an analysis of sequential practices. The range of jobs also cautions against assigning one conventional action or expressive meaning to every click.

Written corpora provide indirect access to this relation. Punctuation and standalone position can indicate detachment, but editorial conventions and genre affect both. Using the same comma as evidence for prosodic isolation and syntactic independence can't then establish an independently observed association between the two. The phonological and grammatical inquiries may converge, but the evidence for their convergence has to preserve the distinction between their measures.

The two phonological inquiries extend the family in different ways. Phonological strata can share their sign-type bearers and enter membership intersections. Prosodic classes of occurrences connect to them through realization and through distributions over contexts. Treating both as extra columns in the same sign-type table would hide the relation that the phonological inquiry is meant to explain.

## 5 Historical ancestry and classes of change

A historical classification asks how signs came to have their present properties. Signs with different current distributions can descend from related sources, while signs with similar current profiles can arrive by different routes. The inquiry can classify present signs by independently established ancestry. It can also classify historical changes, such as clipping or reanalysis. The first has sign-types as bearers; the second has changes linking sources and outcomes.

### 5.1 The documented route to *gee*

Gehweiler (2008) traces *gee* through an invocation of *Jesus*, an exclamatory use of the religious name, and a clipped form. In the invocation, the speaker addresses the deity. The exclamation can express a reaction without that address. Clipping then produces a form whose relation to the name can become opaque. The sequence separates semantic change from the later change of form.

The transitions also have different proposed explanations. Gehweiler relates the first to gradual invited inference, aided by the syntactically peripheral invocation. She analyzes the clipping as an abrupt, taboo-motivated change involving hearer reanalysis (Gehweiler, 2008, pp. 84–88). The source invocation already permits peripheral packaging. Its development gives no reason to assume that a new prosodic boundary caused the semantic transition or that all the properties changed together.

The history explains why the shape of *gee* bears a relation to religious invocation even for a speaker who doesn't recognize that relation. Its present opacity alone wouldn't identify its ancestry. The documented sequence supplies the additional evidence for a restricted instance of path dependence: a particular route helps explain a feature of its outcome, without making every other outcome of that route predictable.

The distinction also corrects an ambiguity in PRIMARY and SECONDARY interjections. Ameka's distinction concerns whether a form has other, non-interjectional uses in the language (Ameka, 1992, pp. 105–106). It isn't a partition into historically underived and historically derived signs. Gehweiler's primary *gee* has a documented derived history. Conversely, an unknown etymology supplies a gap in knowledge, not a common origin shared by all forms whose sources remain unknown.

## 5.2 Source, mechanism, transmission, and trajectory

A useful historical description separates four relations. The SOURCE identifies an earlier sign or construction. The MECHANISM specifies a change, such as clipping or reanalysis. The TRANSMISSION SETTING concerns how the innovation spreads, including borrowing where supported. The subsequent TRAJECTORY concerns further changes in use or form. Signs can share one relation without sharing the other three.

For *gee*, a religious invocation supplies the source and tabooistic clipping supplies one mechanism. Those facts don't establish the transmission history of every later community's use. Nor does a shared source force a common trajectory: an exclamatory religious name can remain phonologically close to its source without later undergoing clipping. Historical classes can be organized at several levels, according to which relation the inquiry needs.

The proposed verbal route to complement-taking *damn* illustrates unequal support among histories. *God damn him* has the shape of an optative, a construction independently recognized in English (Huddleston & Pullum, 2002, p. 944). Omission of the subject and increasing conventional expressive force offer a plausible route to *damn him*. The surviving complement fits that reconstruction. These synchronic comparisons don't establish the sequence as independently as Gehweiler's historical record does.

The source optative already uses plain-form *damn*, so the reconstruction doesn't require loss of tense marking at that transition. And the present complement remains compatible with competing analyses of the head as verbal or interjectional. A classification by independently established verbal ancestry could cross that present grammatical boundary. Inferring the ancestry solely from the complement and then invoking the ancestry to predict that same complement would be circular.

Classes of changes require a different representation. An instance of clipping relates an earlier form to an outcome; an instance of semantic reanalysis relates earlier and later interpretations. Neither is itself a sign-type alongside *gee* and *damn*. A process classification can support comparisons among developments, while explicit source–result relations connect those developments to classifications of their inputs and outputs.

## 5.3 What historical evidence can establish

Historical explanation and forward projection need different comparisons. A documented route may explain a present feature through the changes that produced it. A forward projection asks what information about the route warrants expecting in another case. For a verbal-source class, that target could concern surviving restrictions on dependents, provided those restrictions weren't used to as-

sign the source. The present examples motivate that inquiry without establishing a category-wide result.

Evidence for replication also needs to distinguish innovations from their descendants. Several communities' uses of a borrowed sign can be several outcomes of one innovation, rather than independent instances of a recruitment route. They may support an account of transmission or subsequent divergence. They provide a weaker comparison for the claim that the same source independently produces the same result. The sampling unit depends on which historical claim is at issue.

An ancestry classification can still explain a present contrast even when ancestry adds no predictive information after all its consequences are observed. Conditioning on those consequences changes the question to whether ancestry contributes anything further. That can be a useful test of predictive economy; it isn't a general test of whether the documented process occurred or helped produce the contrast.

Historical classification has two kinds of payoff. It can organize independently supported source relations among present signs, and it can compare processes across developments. These are legitimate contributions without a universal demand for an additional predictive effect. Where a historical category is advertised as projective, its claimed expectations require evidence beyond the histories used to construct it.

## 6 Broader classes and further inquiries

The traditional interjection domain also intersects classes whose most informative members lie elsewhere. Social regulation can group forms by their appropriateness for audiences and settings. Processing inquiries can group familiar expressions by how they're produced or understood. An interjection's membership in either class needn't make the class interjection-specific. Three questions remain distinct: whether the broader class is supported, how it intersects the traditional domain, and whether an INTERJECTION$_X$ label helps describe that intersection.

### 6.1 Social regulation and socially recognized repertoires

Religious exclamations make social regulation relevant to the worked case. In Gehweiler's account, taboo pressure contributes to clipping *Jesus* into *gee*. A present-day inquiry might instead group uses according to their evaluated appropriateness for an audience or setting. The historical cause of an innovation and the current regulation of its descendants are separate targets; evidence for one doesn't settle the other.

Jay and Janschewitz (2008) examine ratings of hypothetical swearing scenarios, finding that judgments depend on the word, speaker–listener relations, and social and physical context. Such evidence supports contextual differentiation in evaluations of taboo language. It doesn't directly observe sanctions or show that corrective responses maintain a class of interjections. Those stronger proposals would need their own measures of behaviour and change.

Social indexing poses a related but distinct inquiry. Agha's account of enregisterment concerns the social recognition of speech forms as indexing attributes of speakers, with a repertoire's associations and social reach subject to change (Agha, 2005, pp. 38–40). A classification by participation in a recognized repertoire could cross grammatical, semantic, and sequential divisions. The relevant evidence concerns recognition and use of those associations, beyond a frequency difference between geographical collections.

These social classifications can matter even where the sequential action stays constant. Two forms might both register surprise while carrying different audience-sensitive evaluations. Conversely, one socially marked repertoire can include forms used for several actions. The intended contribution determines whether to group conventional sign-types, uses in a specified community, or particular selections in context. Where conventions differ by community, that scope belongs in the identification of the bearer.

## 6.2 Formulaicity and processing

Formulaicity offers a similarly broad comparison. A familiar greeting, a pause filler, and an idiom may share aspects of conventionalized production while differing in grammar and conversational function. An inquiry into their processing can group them on grounds that cross the three classifications developed earlier. Its appropriate name may be formulaic expressions, with an explicitly described interjectional intersection.

Van Lancker Sidtis et al. (2015) report differences in formulaic language production and comprehension across clinical groups and healthy participants. Their production analysis combines six kinds of formulaic material, including conversational formulae, pause fillers, sentence stems, and idioms. The results give empirical consequences to that broader organization. They don't isolate an interjection-specific processing class or locate a neural basis for $\textsc{interjection}_{\textsc{syn}}$, $\textsc{interjection}_{\textsc{sem}}$, or $\textsc{interjection}_{\textsc{prag}}$.

An investigation of a narrower class would need an independently motivated grouping and a specified processing contrast. Frequency, familiarity, conventional meaning, and contextual predictability are possible contributors to fluent retrieval. Their roles depend on the question. If repeated use is proposed to help establish the routine, statistical adjustment for measured frequency needs care. The resulting conditional association identifies a causal contribution only under additional assumptions about exposure, retrieval, and their relation to the measure.

Evidence of different processing can also cut across a proposed class. A formulaicity measure might explain performance better than membership in any interjection category. That would favour the broader or continuous representation for this inquiry. The grammatical contrast between a filler and an idiom would remain available to an inquiry into syntax. Local comparative failure has a definite consequence without erasing independently supported distinctions elsewhere.

## 6.3 Keeping the family open without multiplying labels

Other inquiries are easy to motivate, but motivation alone doesn't establish their proposed categories. Table 3 identifies several candidates and the contribution each would need to specify. The table is an agenda for distinguishing questions. It makes no claim that every row identifies a single class, that their extensions are independent, or that all deserve an interjection label.

Morphology illustrates the distinctions within a row. The absence of an inflectional paradigm differs from resistance to derivation. A retained plural shape in exclamatory *heavens!* differs from a productive number contrast. Attested noun and verb uses of *ouch* in child-directed speech show why a claim about invariance needs to specify the conventional use (Stange, 2016, p. 36). These facts motivate several comparisons, without supplying one morphological definition of interjectionhood.

Semiotic and graphic inquiries can likewise cross familiar boundaries. A depiction can be realized as a conventional spoken sign or an inscription, while an inscription can also represent a speaker's production. Work on ideophones distinguishes depictive resources from the predominantly indexical

Table 3: Further inquiries and their possible bearers

| Inquiry | Possible bearer | Contribution requiring assessment |
|---|---|---|
| Morphological organization | Sign-types in identified constructions | Distinguish invariance, productive operations, and retained source material. |
| Semiotic relations | Conventional signs, or sign–referent relations | Organize depiction, indexing, and conventional meaning across grammatical categories. |
| Development and learning | Learners' repertoires or acquisition episodes | Identify acquisition patterns without equating a learner's repertoire with an adult inventory. |
| Multimodal organization | Conventional complexes or situated performances | Explain contributions of vocal and visible components and their coordination. |
| Graphic communication | Written sign-types or particular inscriptions | Distinguish conventional graphic forms from representations of spoken realizations. |
| Annotation | Tokens under a stated coding scheme | Support reliable retrieval or comparison for the scheme's task. |

relation associated with interjections (Dingemanse, 2018; Dingemanse & Akita, 2017). Whether a particular graphic form is a transcription, an autonomous convention, or both requires evidence about its use; resemblance alone doesn't decide.

Developmental and multimodal inquiries can change the bearer more radically. An acquisition episode has participants and a temporal course. A situated vocal–visual performance includes coordination that a lexical sign-type doesn't exhaust. Classifying either can be informative, provided the connection to the sign-type inventory is made through learning or realization relations. Their categories can't be intersected directly with a set of signs as though all were objects of the same kind.

Annotation makes the practical stakes clear. A tag can be useful for retrieving conversational responses even if its extension combines several grammatical or semantic classes. Its success is then measured against the retrieval task and coding policy. That success doesn't establish a lexical category. Conversely, poor retrieval under a particular tag says little about the existence of a distinction the tagging scheme wasn't designed to record.

An inquiry-relative account permits a phonological stratum, an ancestry class, or a formulaic repertoire to keep its informative name. It also permits several such classifications within a single inquiry. What matters is the bearer, the proposed grouping, and the contribution supported by evidence. An open family of classifications needn't become an ever-expanding list of technical senses of INTERJECTION.

## 7 What the corpus evidence bears on

The corpus analyses distinguish useful description from evidence for broader category claims. The Corpus of Global Web-Based English (GloWbE) gives measured geographic distributions; the Corpus of Historical American English (COHA) gives changes in a tagged inventory. Neither operation-

alizes all the classifications above. Two attempted confirmatory analyses also have implementation defects that withdraw their intended interpretations. Keeping these results separate shows how the assessment standard applies when an inquiry has produced numbers but its measurement is incomplete.

## 7.1 Geographic distributions and inventory change

GloWbE contains concentrations of candidate forms in particular country-labelled web sections (Davies & Fuchs, 2015). A model with crossed item and country intercepts estimates a country standard deviation of 0.44, with a 95% credible interval of 0.28–0.65. Its predictive advantage over the model without country variation is positive but imprecise: 17.9 expected log predictive density (ELPD) units, with standard error 13.3; higher values favour the country-conditioned model. Appendix A reports the specification and measurement audit.

The country effect concerns an overall shift in measured frequencies across items. It doesn't establish item-type-specific conditioning, social indexing, or category-boundary differences. Per-million rates combine grammatical availability, preferences, and the frequency of relevant occasions. Country labels identify web sections rather than individual speakers. Raw strings and automatic tags also have different error profiles, and those profiles vary across countries.

The regional particle *lah* illustrates why category assignment requires additional evidence. Lee (2022) analyzes its low-falling variant as adding not-at-issue content concerning the reliability of the grounds for an utterance, with interpretations depending on speech act and context. That analysis is more specific than a broad expressive label. Its not-at-issue status alone doesn't establish the full expressive profile, and regional frequency doesn't settle either the semantic analysis or a conventional sequential role.

The COHA inventory illustrates a different descriptive result (Davies, 2010). In the 1820s, the six most frequent interjection-tagged lemmas are *no*, *oh*, *yes*, *nay*, *o*, and *ah*. In the 2010s, *yeah*, *hey*, and *uh* occupy three of those places. The change concerns frequency ranks: displaced forms needn't disappear, and the tagging policy doesn't independently identify the inquiry-relative categories.

The retained *no*, *oh*, and *yes* share their high ranks across the two periods, but inventory turnover alone doesn't show persistence of a grammatical, semantic, or interactional profile. That would require comparable diagnostics of the relevant uses over time. Figure 1 preserves the descriptive result, whose evidential contribution is the changing recorded inventory.

The frequencies were extracted on 27 July 2026 and archived under `analysis/hand-extracted/`. The 1820s ranking uses the complete 111-form tagged list for that decade, rather than restricting the comparison to the all-period top twenty. This preserves a valid comparison of ranks while leaving the separate question of what the tag includes open to measurement audit.

## 7.2 Withdrawn confirmatory interpretations

The derivative study asked whether verbal uses recruited from *boo*, *ooh*, *shoo*, and *wow* retained source-associated content. The primary script assigned semantic residue to every retained row instead of measuring it independently. Its resulting success counts and intervals reproduce that assignment. They don't test semantic persistence. A separate 97-row recode may supply narrower descriptive evidence if its recorded provenance and coding are sound; it doesn't cover the preregistered early windows adequately (Appendix B).

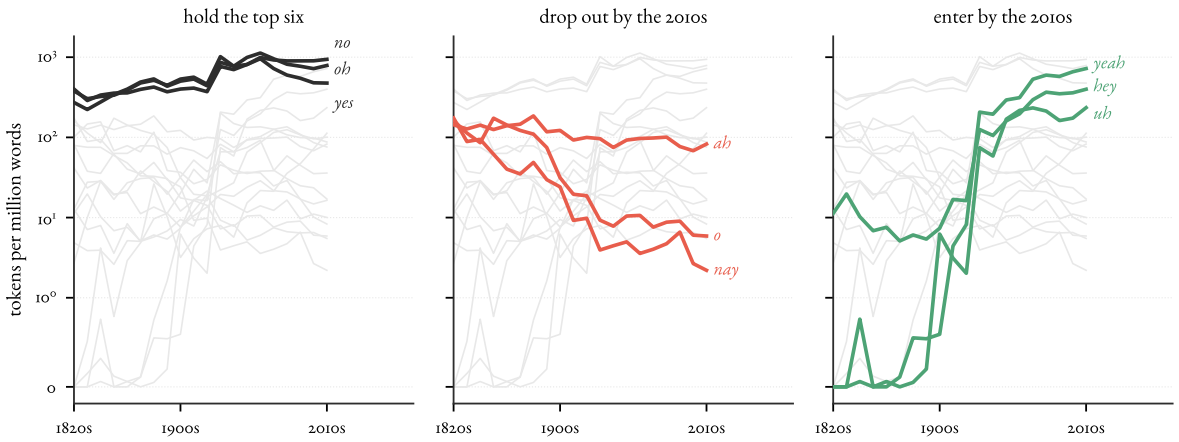



Figure 1: Turnover in the COHA interjection-tagged inventory. Each panel shows the twenty most frequent tagged lemmas across the corpus in grey, with one group highlighted. Three of the 1820s top six are displaced in that ranking by the 2010s. Frequencies per million words use a symmetric log scale retaining zeros. These rankings don't measure stability of a grammatical, semantic, or interactional profile.

Source-associated content would in any event differ from continued expressive-category membership. *The performance wowed us* can describe an impressed response without making the conventional use-conditional contribution of exclamatory *wow!*. A valid study of content continuity would bear on one historical relation. It wouldn't by itself show that a verb retained membership in INTERJECTION$_{SEM}$ while changing its grammatical category.

The *fie* study attempted to compare loss of interactional, semantic, and syntactic properties. Its model input includes obvious optical character recognition (OCR) substitutions for *He*, coded as lacking all three properties. Many other rows have prediction-derived notes without established independent verification. The denominator fails the stated eligibility rule. The archived curves and fitted dates support neither ordered nor simultaneous linguistic erosion (Appendix C).

These failures constrain the claims the worked case can make. The documented history of *gee* remains evidence for that route; the grammatical and semantic contrasts remain available for their respective analyses. The failed corpus tests contribute no independent confirmation of a general pattern of residual meaning or property loss. The framework's distinctions explain the scope of that failure; they don't compensate for it.

## 8 The general consequence

An inquiry-relative account goes beyond observing that different questions invite different classifications. It requires the analyst to say which objects are grouped, which distinctions the grouping preserves, and what evidence supports its intended contribution. For the interjection case, those requirements turn an undifferentiated category dispute into several identifiable comparisons. They also make unsupported proliferation visible.

## 8.1 Pluralism with local failure conditions

Some apparent conflicts disappear when the inquiries are separated. Prenominal *damn* can share expressive meaning with a standalone sign while differing in grammatical distribution. Other conflicts remain within an inquiry: the two analyses of complement-taking *damn* compete over its grammatical organization, and broad versus narrow interactional groupings compete over useful coverage. Pluralism doesn't resolve these disagreements in advance. It identifies the evidence capable of resolving each.

The same discipline supplies failure conditions. A proposed grouping can lack stable membership diagnostics, fail to preserve the claimed contrast, or lose a comparison with a better alternative. A projected property can fail in cases whose membership was established independently. A historical explanation can lack the required source evidence. Changing the inquiry after failure creates a different claim with a different burden; it doesn't preserve the earlier claim's support.

The initial three classifications have no exemption. A broad INTERJECTION$_{\text{PRAG}}$ may organize a domain while its subtypes perform most of the inferential work. A semantic class may prove better expressed as a broader category of expressives. A grammar-facing class may retain unresolved exceptional cases. These are assessable limitations of particular proposals, alongside the descriptive contrasts already supported. The openness of the family permits new candidates under the same conditions.

## 8.2 Overlap, mechanisms, and category standing

Overlap itself supplies neither privilege nor disqualification. For common sign-type bearers, intersections are well-defined and may support useful representation under incomplete observation. Their value depends on the task and alternatives, including representations of the separate memberships and their dependencies. There's no automatic fourth category corresponding to the overlap of grammar, meaning, and interaction, and no reason to prohibit one from proving useful for a specified inquiry.

Categories needn't be statistically independent to do distinct inferential work. A grammatical property can correlate strongly with a prosodic one while each answers a different question. Categories can also share causal mechanisms. One historical process can affect several dimensions; several processes can yield similar outcomes. Evidence for those relations belongs to the causal account, rather than following from the number of categories an analyst recognizes.

A causal network is one possible representation of these dependencies. The representation would need independently identified variables and evidence for the proposed links. Profile stability alone doesn't determine their direction, and convergence alone doesn't show maintenance or corrective control. The inquiry-relative account makes room for such explanations while distinguishing them from the descriptive and representational achievements that can precede them.

Gradience is likewise a separate question. A classification can have graded membership, several sharply defined subtypes, or unresolved cases caused by missing evidence. A prototype account addresses typicality and similarity (Rosch, 1975; Taylor, 2003); an inquiry-relative assessment asks what contribution that organization supports. Either a prototype or a categorical representation can succeed for a specified purpose. The account doesn't make fuzziness a necessary condition of pluralism.

The resulting position permits unequal judgments among classifications without giving a discipline permanent priority. A grouping can be well supported for one question, merely prospective for another, and unhelpful for a third. Its range of success is an empirical and explanatory matter. The

familiar word INTERJECTION marks the historical meeting point of the inquiries, while their justification depends on the work each grouping can support.

## 9 Conclusion

English interjections make a general problem of classification visible. Grammar, expressive meaning, and interaction preserve different relations among conventional signs. Phonological organization and ancestry extend the comparison, while realizations and historical changes require explicit links to sign-type classifications. Broader social and processing classes can be relevant without becoming interjection-specific categories. The family remains open because inquiries can identify further defensible distinctions.

Category standing requires a stated bearer, grouping basis, contribution, scope, and comparison with alternatives. Evidence can support description, projection, economical representation, or causal explanation to different degrees. The worked case supports particular contrasts and a documented recruitment history; its audited corpus attempts don't establish their intended generalizations. This separation allows overlapping classifications to be evaluated without demanding one privileged extension or turning an investigator's purpose into sufficient justification.

### Acknowledgements

I compiled empirical material for the Wikipedia article "English interjections", of which I'm the primary author. This paper reanalyses that material as a worked case for inquiry-relative classification.

The archived data and analysis code are available at https://github.com/BrettRey/interjections-hpc/tree/master. The appendices distinguish their retained descriptive contributions from the confirmatory interpretations withdrawn after audit.

## References


Agha, A. (2005). Voice, footing, enregisterment. *Journal of Linguistic Anthropology*, *15*(1), 38–59. https://doi.org/10.1525/jlin.2005.15.1.38

Ameka, F. (1992). Interjections: The universal yet neglected part of speech. *Journal of Pragmatics*, *18*(2–3), 101–118. https://doi.org/10.1016/0378-2166(92)90048-G

Biber, D., Johansson, S., Leech, G., Conrad, S., & Finegan, E. (1999). *The Longman grammar of spoken and written English*. Longman.

Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–185). MIT Press. https://doi.org/10.7551/mitpress/6396.003.0012

Cram, D. (2008). The exceptional interest of the interjection. *Henry Sweet Society Bulletin*, *50*, 57–66.

Davies, M. (2010). The corpus of historical American English (COHA) [400+ million words, 1820s–2010s]. https://www.english-corpora.org/coha/

Davies, M., & Fuchs, R. (2015). Expanding horizons in the study of World Englishes with the 1.9 billion word Global Web-Based English corpus (GloWbE). *English World-Wide*, *36*(1), 1–28. https://doi.org/10.1075/eww.36.1.01dav

Dingemanse, M. (2018). Redrawing the margins of language: Lessons from research on ideophones. *Glossa: A Journal of General Linguistics*, *3*(1), Article 4, 1–30. https://doi.org/10.5334/gjgl.444

Dingemanse, M. (2024). Interjections at the heart of language. *Annual Review of Linguistics*, *10*, 257–277. https://doi.org/10.1146/annurev-linguistics-031422-124743

Dingemanse, M., & Akita, K. (2017). An inverse relation between expressiveness and grammatical integration: On the morphosyntactic typology of ideophones, with special reference to Japanese. *Journal of Linguistics*, *53*(3), 501–532. https://doi.org/10.1017/S002222671600030X

Dingemanse, M., Torreira, F., & Enfield, N. J. (2013). Is "Huh?" a universal word? conversational infrastructure and the convergent evolution of linguistic items. *PLOS ONE*, *8*(11), Article e78273. https://doi.org/10.1371/journal.pone.0078273

Gehweiler, E. (2008). From proper name to primary interjection: The case of *gee*! *Journal of Historical Pragmatics*, *9*(1), 71–88. https://doi.org/10.1075/jhp.9.1.05geh

Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, *102*(6), 460–465. https://doi.org/10.1511/2014.111.460

Goodman, N. (1955). *Fact, fiction, and forecast*. Harvard University Press.

Huddleston, R., & Pullum, G. K. (2002). *The Cambridge grammar of the English language*. Cambridge University Press. https://doi.org/10.1017/9781316423530

Jay, T., & Janschewitz, K. (2008). The pragmatics of swearing. *Journal of Politeness Research*, *4*(2), 267–288. https://doi.org/10.1515/JPLR.2008.013

Jespersen, O. (1924). *The philosophy of grammar*. George Allen & Unwin.

Lee, J. (2022). An analysis of Colloquial Singapore English *lah* and its interpretation across speech acts. *Languages*, *7*(3), Article 203. https://doi.org/10.3390/languages7030203

Libert, A. (2019). *Interjections*. Oxford University Press, Oxford Bibliographies in Linguistics. https://doi.org/10.1093/obo/9780199772810-0230

Meinard, M. E. M. (2015). Distinguishing onomatopoeias from interjections. *Journal of Pragmatics*, *76*, 150–168. https://doi.org/10.1016/j.pragma.2014.11.011

Ogden, R. (2013). Clicks and percussives in English conversation. *Journal of the International Phonetic Association*, *43*(3), 299–320. https://doi.org/10.1017/S0025100313000224

Potts, C. (2007). The expressive dimension. *Theoretical Linguistics*, *33*(2), 165–198. https://doi.org/10.1515/TL.2007.011

Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1985). *A comprehensive grammar of the English language*. Longman.

Reynolds, B. (2026a). *Difference-in-differences for corpus linguistics: Causal inference, corpus composition, and linguistic change after shocks* [Preprint, LingBuzz/010080]. https://lingbuzz.net/lingbuzz/010080

Reynolds, B. (2026b). *Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds* [Manuscript archived at PhilArchive, 1 June 2026]. https://philarchive.org/rec/REYNES

Reynolds, B. (2026c). *Projectibility: A history and a diagnostic framework* [Manuscript archived at PhilSci-Archive, 22 July 2026, item 30582]. https://philsci-archive.pitt.edu/id/eprint/30582

Rosch, E. (1975). Cognitive representations of semantic categories. *Journal of Experimental Psychology: General*, *104*(3), 192–233. https://doi.org/10.1037/0096-3445.104.3.192

Stange, U. (2016). *Emotive interjections in british english: A corpus-based study on variation in acquisition, function and usage*. John Benjamins. https://doi.org/10.1075/scl.75

Taylor, J. R. (2003). *Linguistic categorization* (3rd ed.). Oxford University Press. https://doi.org/10.1093/oso/9780199266647.001.0001

Unuabonah, F. O., & Daniel, F. O. (2020). Haba! Bilingual interjections in Nigerian English: A corpus-based study. *Journal of Pragmatics*, *163*, 66–77. https://doi.org/10.1016/j.pragma.2020.04.006

Van Lancker Sidtis, D., Choi, J., Alken, A., & Sidtis, J. J. (2015). Formulaic language in Parkinson's disease and Alzheimer's disease: Complementary effects of subcortical and cortical dysfunction. *Journal of Speech, Language, and Hearing Research*, *58*(5), 1493–1507. https://doi.org/10.1044/2015_JSLHR-L-14-0341

Wilkins, D. P. (1992). Interjections as deictics. *Journal of Pragmatics*, *18*(2–3), 119–158. https://doi.org/10.1016/0378-2166(92)90049-H

# A Supplementary materials: GloWbE conditioning analysis

## A.1 Data and measurement

Frequency counts for 18 items were extracted from GloWbE (Davies & Fuchs, 2015): 1.9 billion words across 20 countries. The items are the 10 most frequent interjection-tagged forms (*yes*, *no*, *oh*, *yeah*, *hi*, *hey*, *wow*, *hello*, *ah*, *ha*), three regional forms (*lah*, *yaar*, *haba*), semi-regional *eh*, boundary items *cheers* and *bye*, and fillers *um* and *uh*. Core counts are tag-restricted; regional counts are raw strings because the tagger doesn't recognize the relevant uses. Neither measure identifies the three worked sign-type classifications independently.

Country is assigned to a page rather than a speaker. Assignment uses Google's region function: country domains provide a direct cue, but for `.com` and `.org` sites Google "relies on several heuristics, including 1) the IP address for the web server, 2) who links to that website, and 3) who visits the website" (Davies & Fuchs, 2015, p. 4).

Davies and Fuchs themselves question "how well Google has correctly identified web sites by country". The page list was seeded through Google searches for "hundreds of very high frequency 3-grams (three word strings)" from the Corpus of Contemporary American English (p. 4), anchoring the sampling frame to American frequency data. Country differences may reflect genre mix, internet ecology, and sampling alongside linguistic variation.

Automatic tagging adds another measurement layer. An overall error rate doesn't warrant comparisons if validity varies by country (Reynolds, 2026a). Concordance coding estimates precision, the proportion of tagged hits that are genuine, not recall, the proportion of genuine tokens retrieved. Neither *haba* nor *yaar* receives an interjection tag. All 373 Nigerian *haba* hits are tagged as common or proper nouns, with 65% on a reading including the proper noun. The raw-string regional counts can overstate frequency while a tag-restricted query misses these forms entirely.

## A.2 Geographic distributions

Regional concentration is clear in the raw strings: *lah* occurs at 9.35 per million words in Malaysia and 6.72 in Singapore, and *yaar* at 2.88 in Pakistan and 2.33 in India, with near-zero rates elsewhere. Nigeria's 373 *haba* hits give an upper bound of about 9 per million. Inspecting hits, counting five spellings, and excluding Nigerian Pidgin, indigenous-language material, and duplicates, Unuabonah and Daniel (2020, p. 69) report 248 tokens, or 5.8 per million. The gap illustrates why raw strings and tagged counts aren't directly comparable.

Among the core items, *ah* has the largest cross-country coefficient of variation (CV; standard deviation divided by mean), 0.71: Singapore's 60.42 and Jamaica's 53.16 per million are about eight times Kenya's 7.62. For *yeah*, CV is 0.55 and the United States–Tanzania comparison is approximately 6:1 (103.78 versus 16.85). *Yes* and *no* have lower CVs, 0.21–0.22. The fillers *um* and *uh* have CVs of 0.61 and 0.76, and *bye* has 0.61.

Bootstrap intervals resample the twenty observed country sections. They describe sensitivity to that resampling, not sampling uncertainty over all English-speaking communities: the sections aren't a random sample of such communities. The *yes*/*no* intervals lie below those of eleven of the thirteen other nonregional items, overlapping only *hello* and *ha*. This supports their comparatively low observed dispersion.

The remaining marginal intervals overlap; that doesn't establish equal dispersion or settle pairwise differences. A ranking would require paired contrasts within resamples. Regional concentra-

tion is better represented by the counts: when a form is absent from eighteen sections, resampling can sharply alter the mean and CV. For *lah*, the interval runs from 0.89 to 4.27.

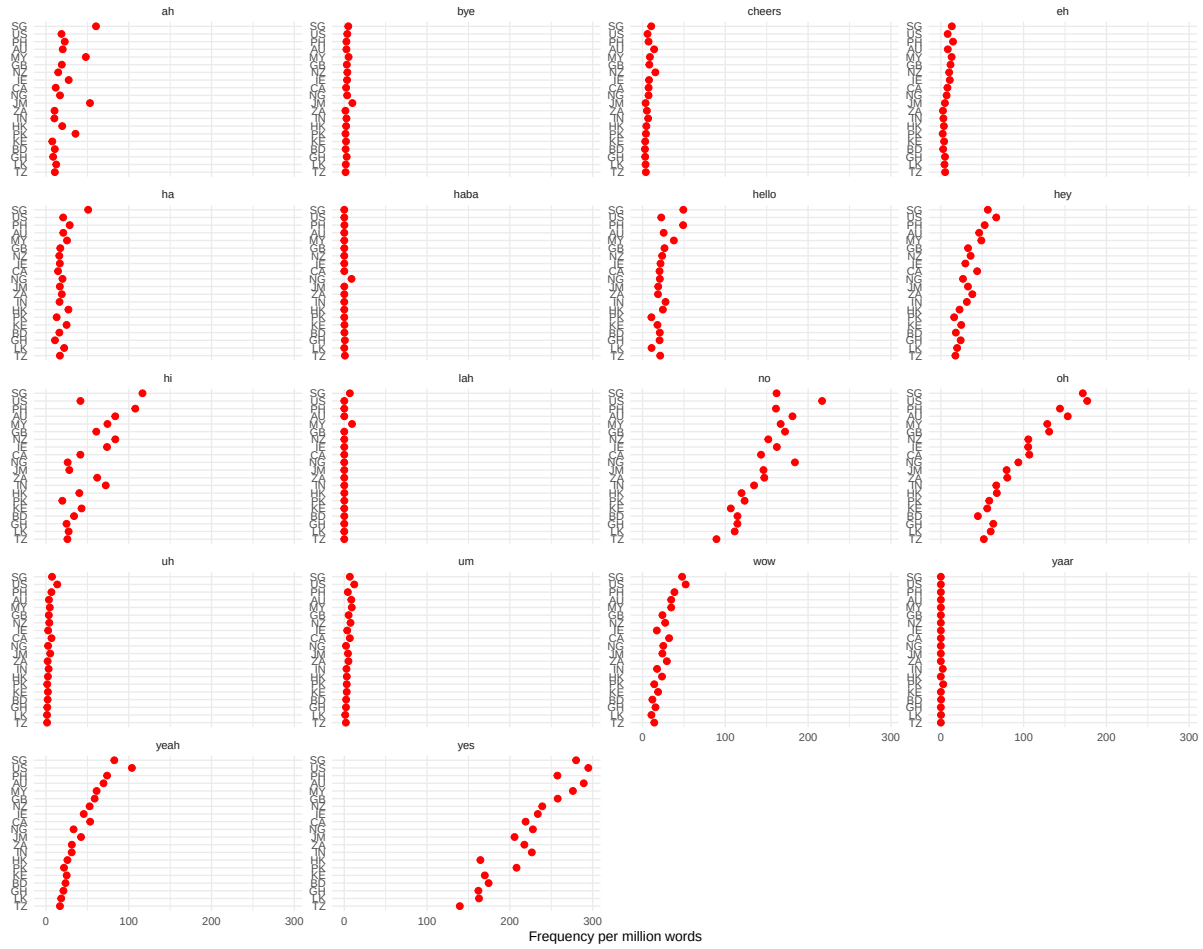



Figure 2: Frequency per million words across 20 GloWbE country sections. Red points give the rate for each country–item combination. Regional strings (*lah*, *yaar*, *haba*) concentrate in one or two countries; *yes* and *no* have comparatively low dispersion. Tagging and raw-string validity differ by item and country, as the audit below documents.

## A.3 Concordance audit and sensitivity

The tagger's training on inner-circle English limits its treatment of regional forms, but recognized items aren't necessarily clean. An initial twenty-token check of tagged *ha* in Kenya and Tanzania retained six and eight interjections respectively; most other hits abbreviated HECTARE in agricultural and development prose. Applying the Kenyan estimate reduces 24.9 tokens per million to about 7.5 and moves Kenya from fifth to nineteenth among the twenty sections. This preliminary estimate differs from the fuller audit below.

The *ha* audit also requires an explicit decision about laughter. Iterated *HA! HA!* may represent laughter, closer to eye dialect than a lexical item, whereas sarcastic or sceptical *ha* is an interjectional

use. Written data make the distinction harder, and digital *haha* may itself be conventionalized. This decision doesn't exclude imitative forms in principle; imitation is one proposed recruitment route. The reported counts include iterated laughter.

Coding tagged concordances across all twenty sections shows the sensitivity. Laughter constitutes 51.5% of tagged *ha* in the United States and 63.0% in Great Britain. Across six defensible treatments of laughter and residual tokens, its CV ranges from 0.40 to 0.83 (cf. Gelman & Loken, 2014); the reported 0.41 includes laughter. The dispersion measure depends on the criterion. A separate ranking result survives: country-specific validity correction moves Kenya from fifth to fifteenth under every coding.

The proportion of tagged *ha* judged interjectional ranges from 34.5% in Bangladesh to 84.0% in Great Britain. Hectare dominates contamination in East Africa and South Asia; personal names matter in Singapore, where only 6.1% of hits are hectare. Pooling country rates to reflect unequal precision – about 8,100 tagged tokens for the United States versus 420 for Ghana – moves no section by more than 0.6 percentage points. Between-section variation exceeds the sampling variation within sections. These checks correct false positives; they don't estimate missed interjections.

Spelling variants introduce another country-dependent error. Raw *ya* and *yeh* look like variants of *yeah*, but they concentrate where substrate words are homographs: *ya* peaks in Tanzania (19.5 per million) and Malaysia (13.9), and *yeh* in Singapore (5.6), Pakistan (4.9), and India (4.9). No *yeh* token carries the interjection tag. Pooling both strings with *yeah* reduces its CV from 0.55 to 0.46 while importing non-English tokens unevenly, so the analysis keeps them separate.

Adding *umm*, *ummm*, and *ummmm* to *um* changes its CV from 0.61 to 0.60 with Pakistan excluded, but to 0.72 with Pakistan included. Pakistan returns nearly eight times as many *umm* as *um* tokens, against a median ratio of about a third elsewhere. In a twenty-token Pakistani sample, nineteen are the Arabic teknonym *Umm*, as in *Umm Salama*, from hadith or biographical material; the twentieth mentions the filler. Raw-string lengthening thus measures substrate homography as well as orthographic variation.

Regional forms face the same problem. Hand-coding sixty-nine *yaar* hits puts the English-matrix rate at roughly 60% in Pakistan and 65% in India; other hits include Tamil *yaar* "who", names, and Roman-script Hindi or Urdu. The relevant twenty-token samples are too coarse for correction, but establish that raw rates are upper bounds. Raw *yes*, *no*, *ha*, *ah*, *oh*, *well*, *right*, and *like* likewise include clausal, quoted, discourse-marker, literary, or noisy uses. The audit establishes a measurement result: validity varies across countries, so pooled frequencies can conceal measurement differences alongside linguistic differences.

## A.4 Model specification

Four Bayesian negative binomial models were fitted in `brms` 2.23 in R, with four chains of 2,000 iterations, including 1,000 warmup iterations. Models m0–m2 use exact reloo for leave-one-out cross-validation; exploratory m3 is assessed through Pareto-smoothed importance sampling (PSIS) diagnostics only.

`m0` (unconditional):

```
count ~ 1 + offset(log(words_m)) + (1 | item)
```

`m1` (conditioned):

```
count ~ 1 + offset(log(words_m)) +
(1 | item) + (1 | country)
```

m2 (item type):

```
count ~ 1 + item_type + offset(log(words_m)) +
(1 | item) + (1 | country)
```

m3 (exploratory):

```
count ~ 1 + item_type + offset(log(words_m)) +
(1 | item) + (1 + item_type || country)
```

The country intercept in m1 estimates an overall frequency shift across items, not an item-specific country effect. Model m3 adds country-varying slopes by item type, but the sparse type cells – semi-regional: 1 item; boundary: 2; filler: 2; regional: 3 – provide little support for those slopes. Neither model estimates grammatical availability separately from preference or frequency of relevant occasions.

## A.5 Results

Exact leave-one-out comparisons for the confirmatory models are:

| Model | ELPD diff | SE diff |
|---|---|---|
| m1 (conditioned) | 0.0 | 0.0 |
| m2 (item type) | −9.3 | 7.9 |
| m0 (unconditional) | −17.9 | 13.3 |

The country-conditioned model has the highest expected log predictive density (ELPD) point estimate, with an imprecise advantage over m0: $\Delta$ELPD = 17.9, standard error 13.3. Adding item type doesn't improve the point estimate. Model m3 isn't formally ranked because nine observations retain Pareto $k > 0.7$ after moment matching, leaving its PSIS estimates unstable.

In m1, the country standard deviation is 0.44 (95% credible interval [CrI]: 0.28–0.65), against an item standard deviation of 2.07 (1.49–2.96), a ratio of 4.7:1. The country interval describes estimated variation under the model; its positive bounds aren't independent evidence against an absent effect, since standard deviations are constrained to be nonnegative. The predictive comparison and descriptive frequencies supply the relevant evidence. With 18 items × 20 countries = 360 observations, the ELPD uncertainty remains substantial.

To propagate the *ha* precision estimates, each country's count was thinned by draws from its BETA posterior, leaving the other seventeen items unchanged, and m0 and m1 were refitted. Within this sensitivity analysis, country standard deviation changes from 0.42 to 0.44 with iterated laughter included and 0.45 with it excluded; the model-comparison values change from 28.8 to 26.9 and 28.5. These approximate leave-one-out values support only an internal comparison, not replacement of the exact-reloo results above. Precision correction leaves recall unidentified.

The evidence supports geographic concentration of particular strings and additional overall country variation in these measured frequencies. It doesn't establish social indexing, item-type-specific conditioning, category-boundary decomposition, or a process maintaining the interjectional profile. The audit explains why those stronger interpretations require independently validated uses and comparable contexts.

# B Supplementary materials: audit of the semantic-residue analysis in COHA

## B.1 Design and extraction

The morphology-gain study was preregistered before confirmatory extraction and coding, after disclosed feasibility queries (`f1a2193`, 27 May 2026). Its fixed COHA source families were *wow*, *boo*, *ooh*, and *shoo*, over 1820–2019. The target derivatives were *wowed*, *wowing*, *wows*; *booed*, *booing*, *boos*; *oohed*, *oohing*, *oohs*; and *shooed*, *shooing*, *shoos*. Bare forms were excluded. Extraction on 27 May 2026 used key-word-in-context (KWIC) exports; chart requests for *oohed* and *oohing* failed, but their KWIC exports loaded. The extraction summary records minor chart/KWIC count discrepancies.

The go/no-go threshold required 20 analyzable derivatives and two decades with at least three each. The archived pipeline marked all families as passing the extraction threshold (GO):

| Family | Included tokens | Decades with 3+ | Status |
|---|---|---|---|
| *boo* | 364 | 10 | GO |
| *ooh* | 52 | 6 | GO |
| *shoo* | 410 | 13 | GO |
| *wow* | 143 | 9 | GO |

## B.2 Intended measures and implementation

The intended confirmatory measures were VP-head syntax and semantic residue: impressed or evaluative response for *wow*, disapproval or contempt for *boo*, appreciative or surprised response for *ooh*, and directive away-sending for *shoo*. Secondary codes concerned interactional residue – recognizable response, uptake, display, or direction – and a recoverable target, object, addressee, or stimulus.

An implementation audit changes the evidential status of the results. The archived primary-coding script first applies exclusion rules, then assigns both `verbal_syntax = 1` and `semantic_residue = 1` to every retained row. The exclusion rules were intended to select verbal uses; their accuracy can't be inferred from the syntax code automatically assigned to every retained row. The resulting file contains 1,462 rows: 969 included and 493 excluded. Semantic persistence wasn't independently measured in this primary coding pass; it was assigned by the pipeline. The outputs below don't constitute a confirmatory result.

Even independently established source content would need separating from expressive meaning type. A verb can describe or cause the response expressed by its source interjection without contributing use-conditionally itself. *The performance wowed us* illustrates that distinction. Continuity of source-associated content would support a limited historical relation, not continued INTERJECTION$_{SEM}$ membership.

## B.3 Archived outputs and their limits

The confirmatory window began with the earliest decade containing at least three included derivatives and extended until it contained ten. Success required both confirmatory codes; the preregistered support rule placed the lower Jeffreys 95% interval bound above 0.5. The pipeline produced:

| Family | Tokens | Window | $n$ | Assigned successes | Jeffreys 95% CI |
|---|---|---|---|---|---|
| *boo* | 364 | 1920s | 17 | 17 | [0.865, 1.000] |
| *ooh* | 52 | 1960s–1980s | 10 | 10 | [0.783, 1.000] |
| *shoo* | 410 | 1890s–1900s | 16 | 16 | [0.857, 1.000] |
| *wow* | 143 | 1930s–1940s | 11 | 11 | [0.800, 1.000] |

All four outputs meet the numerical rule because the semantic outcome was fixed for retained rows. The secondary diachronic model wasn't estimated because the assigned residue variable had no variation. These primary-pipeline records preserve the preregistered attempt; they establish neither semantic persistence nor a general tendency for interjections to acquire morphology or syntax.

### B.4 Reliability record and files

The archived 10% reliability sample contains 97 retained rows (seed 2026). Both comparison files are constant for inclusion, VP-head status, and semantic residue, so their agreement is 1.000 but Cohen's $\kappa$ is undefined. The reliability script instead returned 1.000 when expected agreement equalled one; that output isn't a valid $\kappa$. The sample also excludes rejected rows, so it doesn't audit false exclusions.

The archive contains a separate recorded recode, internally attributed to Brett Reynolds as a blind human pass. Its secondary codes and notes differ from the primary file, but the present implementation audit can't independently verify the recorded human and blind provenance. If that provenance and the coding rubric are sound, the sample can supply narrower descriptive evidence of source-content continuity. Undefined $\kappa$ doesn't erase that possible contribution.

The recode covers only three of the 54 early-window rows used for the confirmatory result: one each for *boo*, *shoo*, and *wow*, and none for *ooh*. It can't replace independent coding of the preregistered windows. The agreement table records the comparison available in the archive, without treating it as validation of the full primary dataset or of the claimed confirmation.

| Code | Agreement | Cohen's $\kappa$ |
|---|---|---|
| Inclusion | 1.000 | undefined |
| VP-head status | 1.000 | undefined |
| Semantic residue | 1.000 | undefined |
| Interactional residue | 0.887 | 0.297 |
| Targeted entity | 0.557 | 0.025 |

The audit uses repository revision `074dfb44cef5b18d292ac176ba5acbfedc76e3b1`. Under `analysis/morphgain-confirmatory/`, the retained outputs are `morphgain-extraction-summary.csv`, `morphgain-go-nogo-summary.csv`, `morphgain-coding-summary.csv`, `morphgain-confirmatory-results.csv`, `morphgain-earliest-windows.csv`, and `morphgain-reliability-results.csv`. The corresponding scripts in `analysis/` are `morphgain-confirmatory-extract.mjs`, `morphgain-confirmatory-prepare.py`, `morphgain-confirmatory-code-primary.py`, `morphgain-confirmatory-analyze.py`, and `morphgain-confirmatory-reliability.py`. They document the implementation; they don't supply the missing independent semantic measurement for the full confirmatory dataset.

# C Supplementary materials: audit of the *fie* analysis in COHA

## C.1 Design, coding, and denominator

The preregistration (e18675b, 15 March 2026) predicted interactional, then semantic, then syntactic loss among attested COHA uses of *fie*, 1820–2019. It excluded metalinguistic mentions, non-English passages, metadata, and optical character recognition (OCR) noise. Represented speech, including fictional dialogue, was eligible. This ordered-erosion hypothesis was an additional historical commitment; it doesn't follow merely from distinguishing three worked sign-type classifications.

The archived analysis contains 616 coded rows. Its exclusion file separately records 141 entries: 134 OCR errors, three metalinguistic mentions, two non-English cases, and two name fragments. However, the model input still includes obvious OCR substitutions coded zero on all three diagnostics, including "FIe stroked his closed eyelids" (F014), "FIe drew her to him" (F028), and "FIe turned his attention" (F043). The model filters only for a non-missing syntactic code. Thus the stated exclusion rule wasn't successfully implemented in the analysed denominator.

The joint syntactic–semantic–interactional codes are 111 for 448 rows, 000 for 139, 011 for 19, 100 for eight, and 010 for two. In addition, 350 rows carry a note beginning "predicted"; the archive doesn't establish independent manual verification of all these assignments. These are implementation findings, not new linguistic classifications. An OCR substitution isn't an interjection that has lost three properties. The archived declines can't establish either ordered or simultaneous linguistic erosion.

## C.2 Archived agreement and proportions

The decade column was hidden in the randomized coding sheet. The reported 10% blind-recode comparison ($n = 62$) yielded:

| Diagnostic | Agreement | Cohen's $\kappa$ |
|---|---|---|
| INTERJECTION$_{\text{SYN}}$ | 87.1% | 0.68 |
| INTERJECTION$_{\text{SEM}}$ | 96.8% | 0.90 |
| INTERJECTION$_{\text{PRAG}}$ | 96.8% | 0.90 |

These archived agreement values don't validate inclusion decisions or resolve the provenance of predicted codes. Figure 3 likewise records the unvalidated input: proportions of 85–100% through the 1890s, a decline between 1900 and 1940, and later fluctuations. Historical fiction and Shakespeare adaptations contribute positive represented uses in the 1980s and 2000s. Publication dates index the sampled texts, not necessarily the contemporary availability of the represented sign-type.

## C.3 Archived models and interpretation

The script fits three separate decade-level binomial logistic regressions, with time centred and scaled in ten-year units. Its normal priors have mean zero and standard deviations 1.5 for the intercept and 0.5 for the slope, with the latter restricted to negative values. Each fit uses four chains, 2,000 warmup and 2,000 subsequent iterations per chain, and seed 2026. The negative-slope restriction imposes monotonic decline and can't model revival. Separate likelihoods also leave the three outcomes' within-token dependence unmodelled.

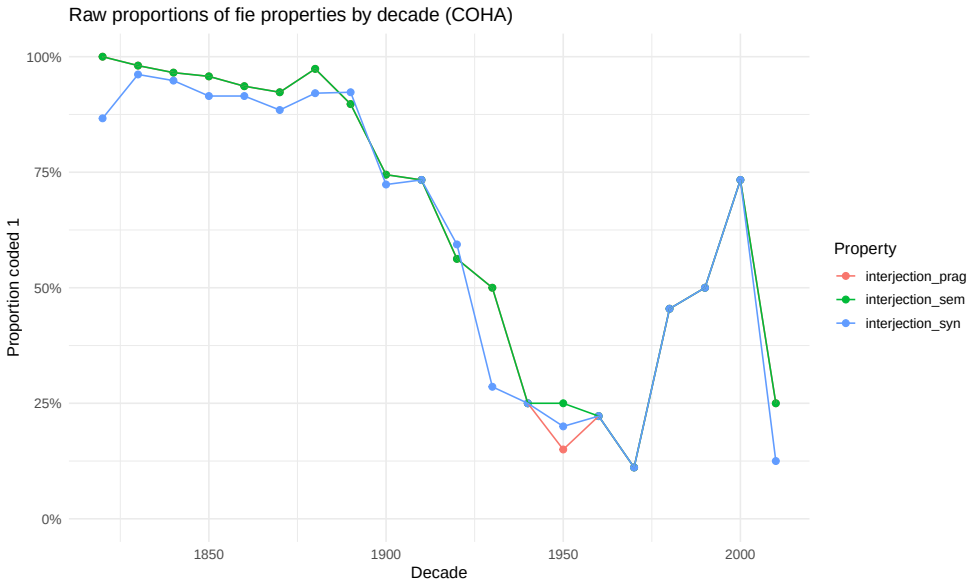



Figure 3: Archived proportions coded 1 by decade in the 616-row *fie* input (COHA, 1820–2019). These are the original diagnostic trajectories, retained to document the attempted analysis. Because OCR substitutions remain in their denominator as zero-valued observations, the curves aren't interpreted as estimates of linguistic property loss.

The script centres decade labels rather than the midpoints specified in the preregistration. Because centring cancels a common offset, this discrepancy leaves the fitted predictor and the ordering probability unchanged. It shifts the conversion back to calendar dates by approximately five years. This bookkeeping issue is separate from the invalid denominator, which blocks the substantive interpretation.

The reported ED50 dates ($\tau$, the 0.5 probability crossing) were:

| Diagnostic | $\tau$ (median) | 95% CrI |
|---|---|---|
| INTERJECTION$_{\text{SYN}}$ | 1947 | 1937–1958 |
| INTERJECTION$_{\text{PRAG}}$ | 1948 | 1939–1959 |
| INTERJECTION$_{\text{SEM}}$ | 1949 | 1941–1960 |

The archived probability of $\tau_{\text{int}} < \tau_{\text{sem}} < \tau_{\text{syn}}$ is 0.13, below the preregistration's informal 0.25 threshold for "inconsistent". Identical independent priors assign each strict ordering probability $1/6$, so 0.13 wouldn't itself be strong evidence against this order even with a valid denominator. Marginal interval overlap wouldn't establish equality or simultaneity either. More fundamentally, the measurement audit supersedes the substantive interpretation of all these outputs: they supply no present test for or against ordered erosion.

The audit is tied to repository revision `074dfb44cef5b18d292ac176ba5acbfedc76e3b1`, especially `fie-coha-preregistration.md`, `analysis/fie-confirmatory-model.R`, `analysis/fie-coding-sheet-annotated.csv`, and `analysis/fie-excluded.csv`. A valid comparison would first distinguish eligible occurrences from noise, then separate changing mixtures of sign-types from diagnostic change conditional on a bearer. No recoding or replacement model is reported here.

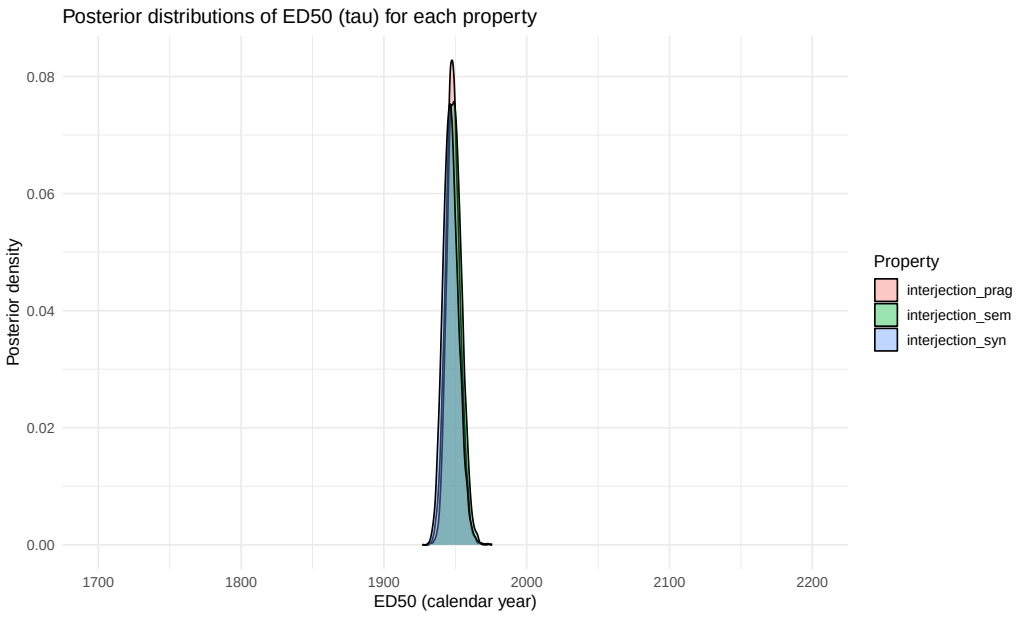



Figure 4: Archived ED50 distributions from an analysis whose input failed the preregistered exclusion rule. The numerical outputs are retained for the audit record; their overlap isn't evidence for simultaneous loss, and the distributions aren't interpreted as valid estimates of linguistic erosion.